

# Measures and Integration: An Informal Introduction

For many students who are learning measure and integration theory for the first time, the notions of a  $\sigma$ -algebra of subsets of a set  $\Omega$ , countable additivity of a set function  $\lambda$ , measurability of a function, the definition of an integral, and the interchange of limits and integration are not easy to understand and often seem not so intuitive. The goals of this informal introduction to this subject are (1) to show that the notions of  $\sigma$ -algebra and countable additivity are logical consequences of certain natural approximation procedures; (2) the dividends for the assumption of these two properties are great, and they lead to a nice and natural theory that is also very powerful for the handling of limits. Of course, as the saying goes, the devil is in the details. After this informal introduction, the necessary details are given in the next few sections. It is hoped that after this heuristic explanation of the subject, the motivation for and the process of mastering the details on the part of the students will be forthcoming.

## What is Measure Theory?

A simple answer is that it is a theory about the distribution of mass over a set  $S$ . If the mass is uniformly distributed and  $S$  is an Euclidean space  $\mathbb{R}^k$ , it is the theory of Lebesgue measure on  $\mathbb{R}^k$  (i.e., length in  $\mathbb{R}$ , area in  $\mathbb{R}^2$ , volume in  $\mathbb{R}^3$ , etc.). Probability theory is concerned with the case when  $S$  is the sample space of a random experiment and the total mass is one. Consider the following example.

Imagine an open field  $S$  and a snowy night. At daybreak one goes to the field to measure the amount of snow in as many of the subsets of  $S$  as

possible. Suppose now that one has the tools to measure the snow exactly on a class of subsets, such as triangles, rectangles, circular shapes, elliptic shapes, etc., no matter how small. It is natural to try to approximate oddly-shaped regions by combinations of these “standard shapes,” and then use a limiting process to obtain a measure for the oddly-shaped regions and reach some limit for such sets. Let  $\mathcal{B}$  denote the class of subsets of  $\mathbb{S}$  whose measure is obtained this way and let  $\lambda(B)$  denote the amount of snow in each  $B \in \mathcal{B}$ . Call  $\mathcal{B}$  the class of all (snow) measurable sets and  $\lambda(B)$  the measure (of snow) on  $B$  for each  $B \in \mathcal{B}$ . It is reasonable to expect that the following properties of  $\mathcal{B}$  and  $\lambda(\cdot)$  hold:

### Properties of $\mathcal{B}$

- (i)  $A \in \mathcal{B} \Rightarrow A^c \in \mathcal{B}$  (i.e., if one can measure the amount of snow on  $A$  and knows the total amount on  $\mathbb{S}$ , then one knows the amount of snow on  $A^c$ ).
- (ii)  $A_1, A_2 \in \mathcal{B} \Rightarrow A_1 \cup A_2 \in \mathcal{B}$  (i.e., if one can measure the amount of snow on  $A_1$  and  $A_2$ , then one can do the same for  $A_1 \cup A_2$ ).
- (iii) If  $\{A_n : n \geq 1\} \subset \mathcal{B}$ , and  $A_n \subset A_{n+1}$  for all  $n \geq 1$ , then  $\lim_{n \rightarrow \infty} A_n \equiv \bigcup_{n=1}^{\infty} A_n \in \mathcal{B}$  (i.e., if one can measure the amount of snow on  $A_n$  for each  $n \geq 1$  on an *increasing sequence* of sets, then one can do so on the limit of  $A_n$ ).
- (iv)  $\mathcal{C} \subset \mathcal{B}$  where  $\mathcal{C}$  is the class of nice sets such as triangles, squares, etc., that one started with.

### Properties of $\lambda(\cdot)$

- (i)  $\lambda(A) \geq 0$  for  $A \in \mathcal{B}$  (i.e., the amount of snow on any set is nonnegative!)
- (ii) If  $A_1, A_2 \in \mathcal{B}$ ,  $A_1 \cap A_2 = \emptyset$ ,  $\lambda(A_1 \cup A_2) = \lambda(A_1) + \lambda(A_2)$  (i.e., the amounts of snow on two disjoint sets simply add up! This property of  $\lambda$  is referred to as *finite additivity*).
- (iii) If  $\{A_n : n \geq 1\} \subset \mathcal{B}$ , are such that  $A_n \subset A_{n+1}$  for all  $n$ , then  $\lambda(\lim_{n \rightarrow \infty} A_n) = \lambda(\bigcup_{n=1}^{\infty} A_n) = \lim_{n \rightarrow \infty} \lambda(A_n)$  (i.e., if we can approximate a set  $A$  by an increase sequence of sets  $\{A_n\}_{n \geq 1}$  from  $\mathcal{B}$ , then  $\lambda(A) = \lim_{n \rightarrow \infty} \lambda(A_n)$ . This property of  $\lambda$  is referred to as *monotone continuity from below*, or *m.c.f.b.* in short).

This last assumption (iii) is what guarantees that different approximations lead to consistent limits. Thus, if there are two increasing sequences  $\{A'_n\}_{n \geq 1}$  and  $\{A''_n\}_{n \geq 1}$  having the same limit  $A$  but  $\{\lambda(A'_n)\}_{n \geq 1}$  and  $\{\lambda(A''_n)\}_{n \geq 1}$  have different limits, then the approximating procedures are not consistent.

It turns out that the above set of reasonable and natural assumptions lead to a very rich and powerful theory that is widely applicable.

A triplet  $(\mathbb{S}, \mathcal{B}, \lambda)$  that satisfies the above two sets of assumptions is called a *measure space*. The assumptions on  $\mathcal{B}$  and  $\lambda$  are *equivalent* to the following:

**On  $\mathcal{B}$**

$\mathcal{B}(i)'$ :  $\emptyset$ , the empty set, lies in  $\mathcal{B}$

$\mathcal{B}(ii)'$ :  $A \in \mathcal{B} \Rightarrow A^c \in \mathcal{B}$  (same as (i) before)

$\mathcal{B}(iii)'$ :  $A_1, A_2, \dots \in \mathcal{B} \Rightarrow \cup_i A_i \in \mathcal{B}$  (combines (ii) and (iii) above) (*Closure under countable unions*).

**On  $\lambda$**

$\lambda(i)'$ :  $\lambda(\cdot) \geq 0$  (same as (i) before) and  $\lambda(\emptyset) = 0$ .

$\lambda(ii)'$ :  $\lambda(\cup_{n \geq 1} A_n) = \sum_{n=1}^{\infty} \lambda(A_n)$  if  $\{A_n\}_{n \geq 1} \subset \mathcal{B}$  are *pairwise disjoint*, i.e.,  $A_i \cap A_j = \emptyset$  for  $i \neq j$  (*Countable additivity*).

Any collection  $\mathcal{B}$  of subsets of  $\mathbb{S}$  that satisfies  $\mathcal{B}(i)'$ ,  $\mathcal{B}(ii)'$ ,  $\mathcal{B}(iii)'$  above is called a  $\sigma$ -*algebra*. Any set function  $\lambda$  on a  $\sigma$ -algebra  $\mathcal{B}$  that satisfies  $\lambda(i)'$  and  $\lambda(ii)'$  above is called a *measure*. Thus, a *measure space* is a triplet  $(\mathbb{S}, \mathcal{B}, \lambda)$  where  $\mathbb{S}$  is a nonempty set,  $\mathcal{B}$  is a  $\sigma$ -algebra of subsets of  $\mathbb{S}$  and  $\lambda$  is a *measure* on  $\mathcal{B}$ . Notice that the  $\sigma$ -algebra structure on  $\mathcal{B}$  and the countable additivity of  $\lambda$  are necessary consequences of the very natural assumptions (i), (ii), and (iii) on  $\mathcal{B}$  and  $\lambda$  defined at the beginning.

It is not often the case that one is given  $\mathcal{B}$  and  $\lambda$  explicitly. Typically, one starts with a small collection  $\mathcal{C}$  of subsets of  $\mathbb{S}$  that have properties resembling intervals or rectangles and a set function  $\lambda$  on  $\mathcal{C}$ . Then,  $\mathcal{B}$  is the smallest  $\sigma$ -algebra containing  $\mathcal{C}$  obtained from  $\mathcal{C}$  by various operations such as countable unions, intersections, and their limits. The key properties on  $\mathcal{C}$  that one needs are:

(i)  $A, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C}$  (e.g., intersection of intervals is an interval).

(ii)  $A \in \mathcal{C} \Rightarrow A^c$  is a finite union of sets from  $\mathcal{C}$  (e.g., the complement of an interval is the union of two intervals or an interval itself).

A collection  $\mathcal{C}$  satisfying (i) and (ii) is called a *semialgebra*. The function  $\lambda$  on  $\mathcal{B}$  is an extension of  $\lambda$  on  $\mathcal{C}$ . For this extension to be a measure on  $\mathcal{B}$ , the conditions needed are

(i)  $\lambda(A) \geq 0$  for all  $A \in \mathcal{C}$

(ii) If  $A_1, A_2, \dots \in \mathcal{C}$  are pairwise disjoint and  $A = \bigcup_{n \geq 1} A_n \in \mathcal{C}$ , then  $\lambda(A) = \sum_{n=1}^{\infty} \lambda(A_n)$ .

There is a result, known as the *extension theorem*, that says that given such a pair  $(\mathcal{C}, \lambda)$ , it is possible to extend  $\lambda$  to  $\mathcal{B}$ , the smallest  $\sigma$ -algebra containing  $\mathcal{C}$ , such that  $(\mathbb{S}, \mathcal{B}, \lambda)$  is a measure space. Actually, it does more. It constructs a  $\sigma$ -algebra  $\mathcal{B}^*$  larger than  $\mathcal{B}$  and a measure  $\lambda^*$  on  $\mathcal{B}^*$  such that  $(\mathbb{S}, \mathcal{B}^*, \lambda^*)$  is a larger measure space,  $\lambda^*$  coincides with  $\lambda$  on  $\mathcal{C}$  and it provides nice approximation theorems. For example, the following approximation result is available:

If  $B \in \mathcal{B}^*$  with  $\lambda^*(B) < \infty$ , then for every  $\epsilon > 0$ ,  $B$  can be approximated by a finite union of sets from  $\mathcal{C}$ , i.e., there exist sets  $A_1, \dots, A_k \in \mathcal{C}$  with  $k < \infty$  such that  $\lambda^*(A \triangle B) < \epsilon$  where  $A \equiv \bigcup_{i=1}^k A_i$  and  $A \triangle B = (A \cap B^c) \cup (A^c \cap B)$ , the *symmetric difference* between  $A$  and  $B$ .

That is, in principle, every (measurable) set  $B$  of finite measure (i.e.,  $B$  belonging to  $\mathcal{B}^*$  with  $\lambda^*(B) < \infty$ ) is nearly a finite union of (elementary) sets that belong to  $\mathcal{C}$ . For example if  $\mathbb{S} = \mathbb{R}$  and  $\mathcal{C}$  is the class of intervals, then every measurable set of finite measure is nearly a finite union of disjoint bounded open intervals.

The following are some concrete examples of the above extension procedure.

**Theorem:** (*Lebesgue-Stieltjes measures on  $\mathbb{R}$* ). Let  $F : \mathbb{R} \rightarrow \mathbb{R}$  satisfy

- (i)  $x_1 < x_2 \Rightarrow F(x_1) \leq F(x_2)$  (*nondecreasing*);
- (ii)  $F(x) = F(x+) \equiv \lim_{y \downarrow x} F(y)$  for all  $x \in \mathbb{R}$  (i.e.,  $F(\cdot)$  is *right continuous*).

Let  $\mathcal{C}$  be the class of sets of the form  $(a, b]$ , or  $(b, \infty)$ ,  $-\infty \leq a < b < \infty$ . Then, there exists a measure  $\mu_F$  defined on  $\mathcal{B} \equiv \mathcal{B}(\mathbb{R})$ , the smallest  $\sigma$ -algebra generated by  $\mathcal{C}$  such that

$$\mu_F((a, b]) = F(b) - F(a) \quad \text{for all } -\infty < a < b < \infty.$$

The  $\sigma$ -algebra  $\mathcal{B} \equiv \mathcal{B}(\mathbb{R})$  is called the *Borel  $\sigma$ -algebra of  $\mathbb{R}$* .

**Corollary:** *There exists a measure  $m$  on  $\mathcal{B}(\mathbb{R})$  such that  $m(I) = \text{the length of } I$ , for any interval  $I$ .*

**Proof:** Take  $F(x) \equiv x$  in the above theorem.

This measure is called the *Lebesgue measure on  $\mathbb{R}$* . □

**Corollary:** *There exists a measure  $\lambda$  on  $\mathcal{B}(\mathbb{R})$  such that*

$$\lambda((a, b]) = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-x^2/2} dx.$$

**Proof:** Take  $F(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du$ ,  $x \in \mathbb{R}$ .

This measure is called the *standard normal probability measure* on  $\mathbb{R}$ .  $\square$

**Theorem:** (*Lebesgue-Stieltjes measures on  $\mathbb{R}^2$* ). Let  $F : \mathbb{R}^2 \rightarrow \mathbb{R}$  be a function satisfying the following:

- (i) (*Monotonicity*) For  $x = (x_1, x_2)', y = (y_1, y_2)'$  with  $x_i \leq y_i$  for  $i = 1, 2$ ,  $(\Delta F)(x, y) \equiv F(y_1, y_2) - F(x_1, y_2) - F(y_1, x_2) + F(x_1, x_2) \geq 0$ .
- (ii) (*Continuity from above*)  $F(x) = \lim_{y_i \downarrow x_i, i=1,2} F(y)$  for all  $x \in \mathbb{R}^2$ .

Let  $\mathcal{C}$  be the class of all rectangles of the form  $(a, b] \equiv (a_1, b_1] \times (a_2, b_2]$  with  $a = (a_1, a_2)', b = (b_1, b_2)' \in \mathbb{R}^2$ . Then there exists a measure  $\mu_F$ , defined on the  $\sigma$ -algebra  $\mathcal{B} \equiv \mathcal{B}(\mathbb{R}^2)$ , generated by  $\mathcal{C}$ , such that

$$\mu_F((a, b]) = (\Delta F)(a, b).$$

The above theorems have a converse that says that every measure on  $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$  that is finite on bounded sets arises from some function  $F$  (called a *distribution function*) and is, therefore, a *Lebesgue-Stieltjes measure*.

Here is another simple example of a measure space (with discrete  $\mathbb{S}$ ).

**Example:** Let  $\mathbb{S} = \{s_1, s_2, \dots, s_k\}, k \leq \infty$ , and let  $\mathcal{B} = \mathcal{P}(\mathbb{S})$ , the power set of  $\mathbb{S}$ , i.e., the collection of all possible subsets of  $\mathbb{S}$ . Let  $p_1, p_2, \dots$  be nonnegative numbers. Let

$$\lambda(A) \equiv \sum_{1 \leq i \leq k} p_i I_A(s_i),$$

where  $I_A$  is the indicator function of the set  $A$ , defined by  $I_A(s) = 1$  if  $s \in A$  and 0 otherwise. It is easy to verify that  $(\mathbb{S}, \mathcal{B}, \lambda)$  is a measure space and also that every measure  $\lambda$  on  $\mathcal{B}$  arises this way.

### What is Integration Theory?

In short, it is a theory about weighted sums of functions on a set  $\mathbb{S}$  when the weights are specified by a mass distribution  $\lambda$ . Here is a more detailed answer.

Let  $(\mathbb{S}, \mathcal{B}, \lambda)$  be a measure space. Suppose  $f : \mathbb{S} \rightarrow \mathbb{R}$  is a *simple function*, i.e.,  $f$  is such that  $f(\mathbb{S})$  is a finite set  $\{a_1, a_2, \dots, a_k\}$ . It is reasonable to define the weighted sum of  $f$  with respect to  $\lambda$  as  $\sum_{i=1}^k a_i \lambda(A_i)$  where  $A_i = f^{-1}\{a_i\}$ . Of course, for this to be well defined, one needs  $A_i$  to be in  $\mathcal{B}$  and  $\lambda(A_i) < \infty$  for all  $i$  such that  $a_i \neq 0$ .

Notice that the quantity  $\sum_{i=1}^k a_i \lambda(A_i)$  remains the same whether the  $a_i$ 's are distinct or not. Call this the integral of  $f$  with respect to  $\lambda$  and denote

this by  $\int f d\lambda$ . If  $f$  and  $g$  are simple, then for  $\alpha, \beta \in \mathbb{R}$ ,  $\int(\alpha f + \beta g)d\lambda = \alpha \int f d\lambda + \beta \int g d\lambda$ . Now how should one define  $\int f d\lambda$  (integral of  $f$  with respect to  $\lambda$ ) for a nonsimple  $f$ ? The answer, of course, is to “approximate” by simple functions. Let  $f$  be a nonnegative function. To define the integral of  $f$ , one would like to approximate  $f$  by *simple* functions. It turns out that a necessary and sufficient condition for this is that for any  $a \in \mathbb{R}$ , the set  $\{s : f(s) \leq a\}$  is in  $\mathcal{B}$ . Such a function  $f$  is called *measurable* with respect to  $\mathcal{B}$  or  *$\mathcal{B}$ -measurable* or simply, *measurable* (if  $\mathcal{B}$  is kept fixed throughout). Let  $f$  be a nonnegative  $\mathcal{B}$  measurable function. Then there exists a sequence  $\{f_n\}_{n \geq 1}$  of *simple nonnegative* functions such that for each  $s \in \mathbb{S}$ ,  $\{f_n(s)\}_{n \geq 1}$  is a nondecreasing sequence converging to  $f(s)$ . It is now natural to define the weighted sum of  $f$  with respect to  $\lambda$ , i.e., the *integral of  $f$  with respect to  $\lambda$* , denoted by  $\int f d\lambda$ , as

$$\int f d\lambda = \lim_{n \rightarrow \infty} \int f_n d\lambda.$$

An immediate question is: *Is the right side the same for all such approximating sequences  $\{f_n\}_{n \geq 1}$ ?* The answer is a *yes*; it is guaranteed by the very natural assumption imposed on  $\lambda$  that it is finitely additive and monotone continuous from below, i.e.  $\lambda(\text{ii})$  and  $\lambda(\text{iii})$  (or equivalently, that  $\lambda$  is countably additive, i.e.,  $\lambda(\text{ii})'$ ).

One can strengthen this to a stronger result known as the *monotone convergence theorem*, a key result that in turn leads to two other major convergence results.

**The monotone convergence theorem (MCT):** *Let  $(\mathbb{S}, \mathcal{B}, \lambda)$  be a measure space and let  $f_n : \mathbb{S} \rightarrow \mathbb{R}_+$ ,  $n \geq 1$  be a sequence of nonnegative  $\mathcal{B}$ -measurable functions (not necessarily simple) such that for all  $s \in \mathbb{S}$ ,*

$$(i) \quad f_n(s) \leq f_{n+1}(s), \quad \text{for all } n \geq 1, \quad \text{and}$$

$$(ii) \quad \lim_{n \rightarrow \infty} f_n(s) = f(s).$$

*Then  $f$  is  $\mathcal{B}$ -measurable and  $\int f d\lambda = \lim_{n \rightarrow \infty} \int f_n d\lambda$ .*

This says that the integral and the limit can be interchanged for monotone nondecreasing nonnegative  $\mathcal{B}$ -measurable functions. Note that if  $f_n = I_{A_n}$ , the indicator function of a set  $A_n$  and if  $A_n \subset A_{n+1}$  for each  $n$ , then the MCT is the same as m.c.f.b. (cf. property  $\lambda(\text{ii})$ ). *Thus, the very natural assumption of m.c.f.b. yields a basic convergence result that makes the integration theory so elegant and powerful.*

To extend the definition of  $\int f d\lambda$  to a real valued,  $\mathcal{B}$ -measurable function  $f : \mathbb{S} \rightarrow \mathbb{R}$ , one uses the simple idea that  $f$  can be decomposed as  $f = f^+ - f^-$  where  $f^+(s) = \max\{f(s), 0\}$  and  $f^-(s) = \max\{-f(s), 0\}$ ,  $s \in \mathbb{S}$ . Since both  $f^+$  and  $f^-$  are nonnegative and  $\mathcal{B}$ -measurable,  $\int f^+ d\lambda$  and

$\int f^- d\lambda$  are both well defined. Now set

$$\int f d\lambda = \int f^+ d\lambda - \int f^- d\lambda,$$

provided at least one of the two terms on the right is finite. The function  $f$  is said to be *integrable* with respect to (w.r.t.)  $\lambda$  if both  $\int f^+ d\lambda$  and  $\int f^- d\lambda$  are finite or, equivalently, if  $\int |f| d\lambda < \infty$ . The following is a consequence of the MCT.

**Fatou's lemma:** *Let  $\{f_n\}_{n \geq 1}$  be a sequence of nonnegative  $\mathcal{B}$ -measurable functions on a measure space  $(\mathbb{S}, \mathcal{B}, \lambda)$ . Then*

$$\int \liminf_{n \rightarrow \infty} f_n d\lambda \leq \liminf_{n \rightarrow \infty} \int f_n d\lambda.$$

This in turn leads to

**(Lebesgue's) dominated convergence theorem (DCT):** *Let  $\{f_n\}_{n \geq 1}$  be a sequence of  $\mathcal{B}$ -measurable functions from a measure space  $(\mathbb{S}, \mathcal{B}, \lambda)$  to  $\mathbb{R}$  and let  $g$  be a  $\mathcal{B}$ -measurable nonnegative integrable function on  $(\mathbb{S}, \mathcal{B}, \lambda)$ . Suppose that for each  $s$  in  $\mathbb{S}$ ,*

$$(i) \quad |f_n(s)| \leq g(s) \quad \text{for all } n \geq 1 \quad \text{and}$$

$$(ii) \quad \lim_{n \rightarrow \infty} f_n(s) = f(s).$$

*Then,  $f$  is integrable and*

$$\lim_{n \rightarrow \infty} \int f_n d\lambda = \int f d\lambda = \int \lim_{n \rightarrow \infty} f_n d\lambda.$$

Thus some very natural assumptions on  $\mathcal{B}$  and  $\lambda$  lead to an interesting measure and integration theory that is quite general and that allows the interchange of limits and integrals under fairly general conditions. A systematic treatment of the measure and integration theory is given in the next five chapters.